

Protein-Protein Inhibitors of RAC1-GEF (Prex1)

**New directions to the Lead Discovery/Optimization RAC1
modulators**

EM & ST

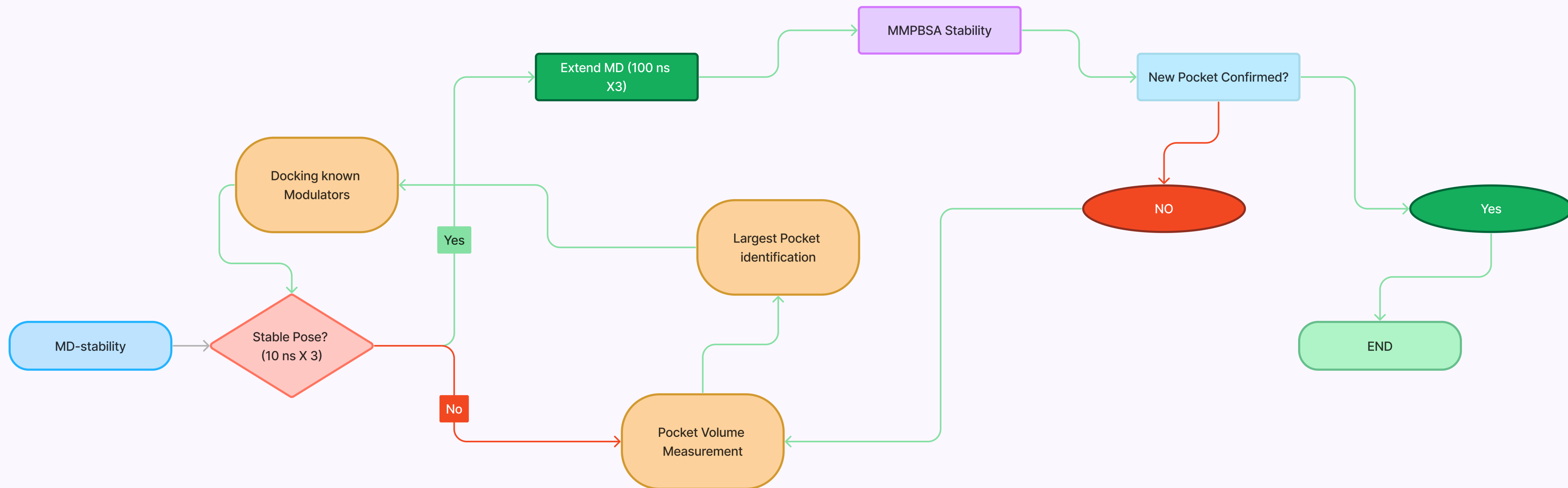
MD-based Binding Affinity Calculations

- The RAC1 affinity to Prex1 was computed as a measure to observe the stability of RAC1-Prex1 complex.
- The affinity of unliganded RAC1 to Prex1 was found to be **-49.66(9.2)**.
- The ddG(RAC1) was computed by subtracting the unliganded affinity to the liganded affinity.

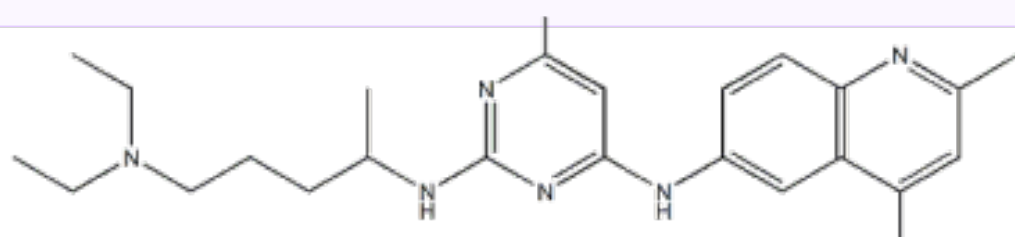
Molecule	Ligand Affinity	RAC1 Affinity	ddG (RAC1)
A4.1	-50.75(5.5)	-56.18(14.5)	6.52
A457	-45.17(6.9)	-57.01(10.37)	7.35
West-R5	-45.34(5.6)	-58.60(8.8)	8.94
ITX-30470	-36.61(8.99)	-61.56(10.62)	11.9
ITX-30450	-17.26(7.65)	-59.6(11.35)	9.94

Work Flow for manuscript

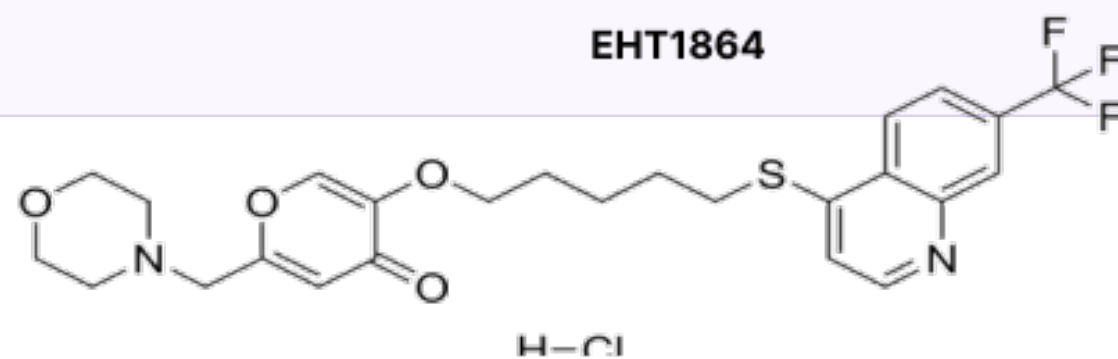
Work Flow For Pocket Confirmation



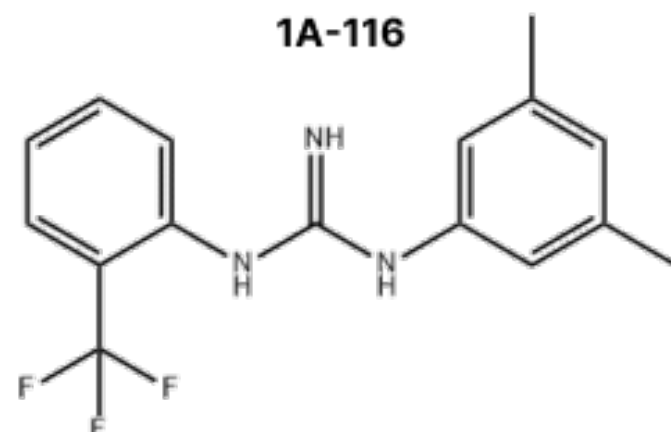
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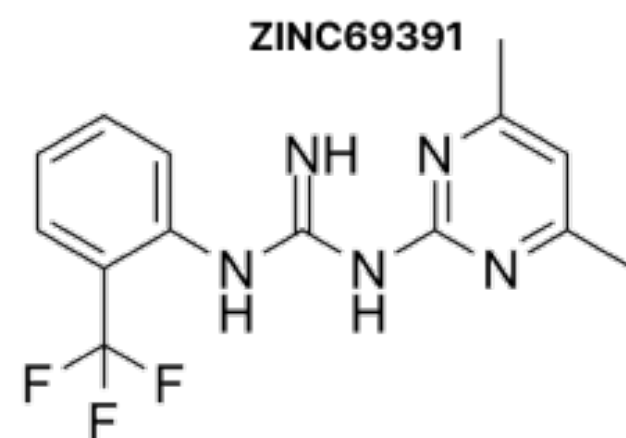
EHT1864



1A-116



ZINC69391

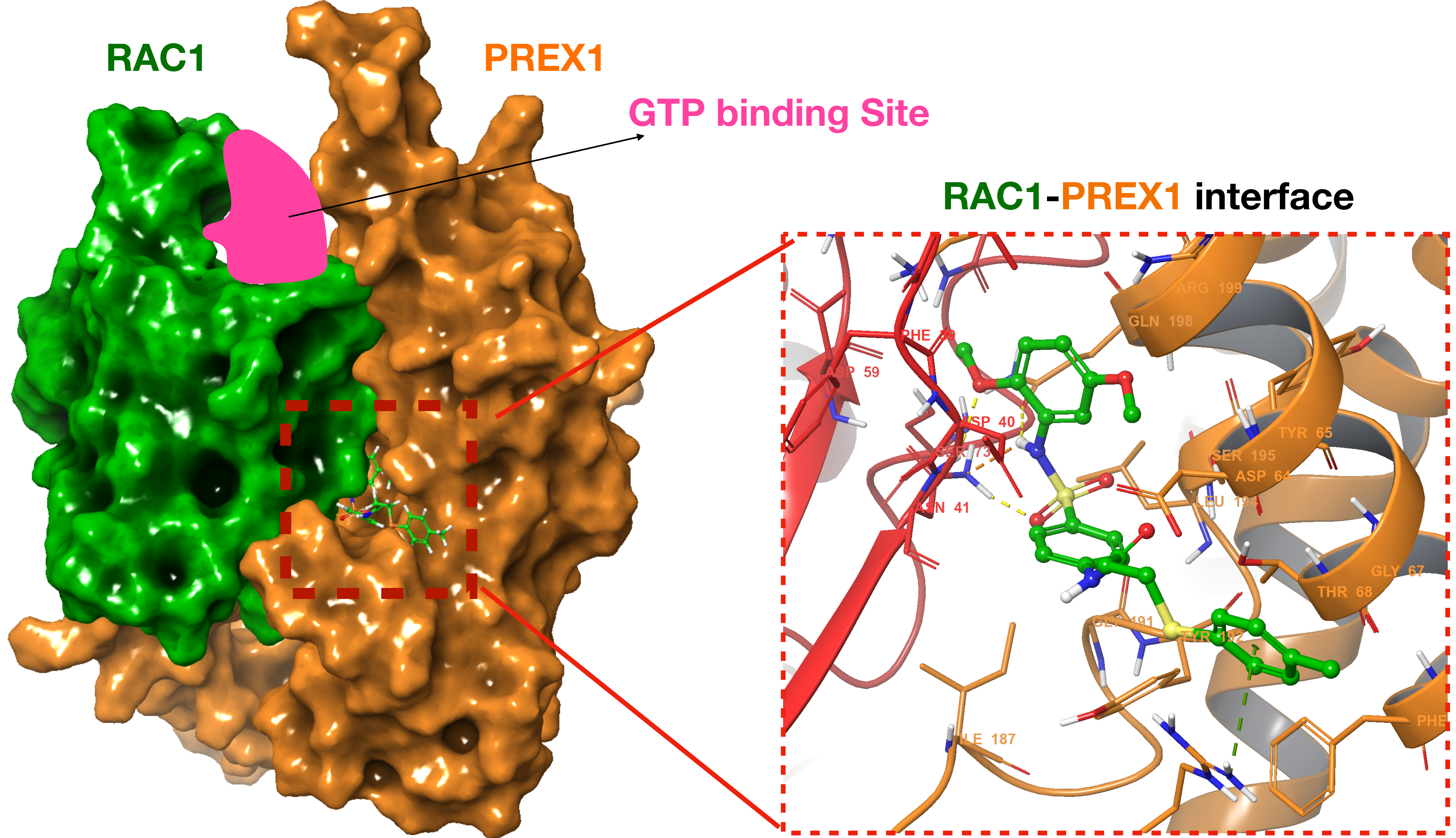


RAC1

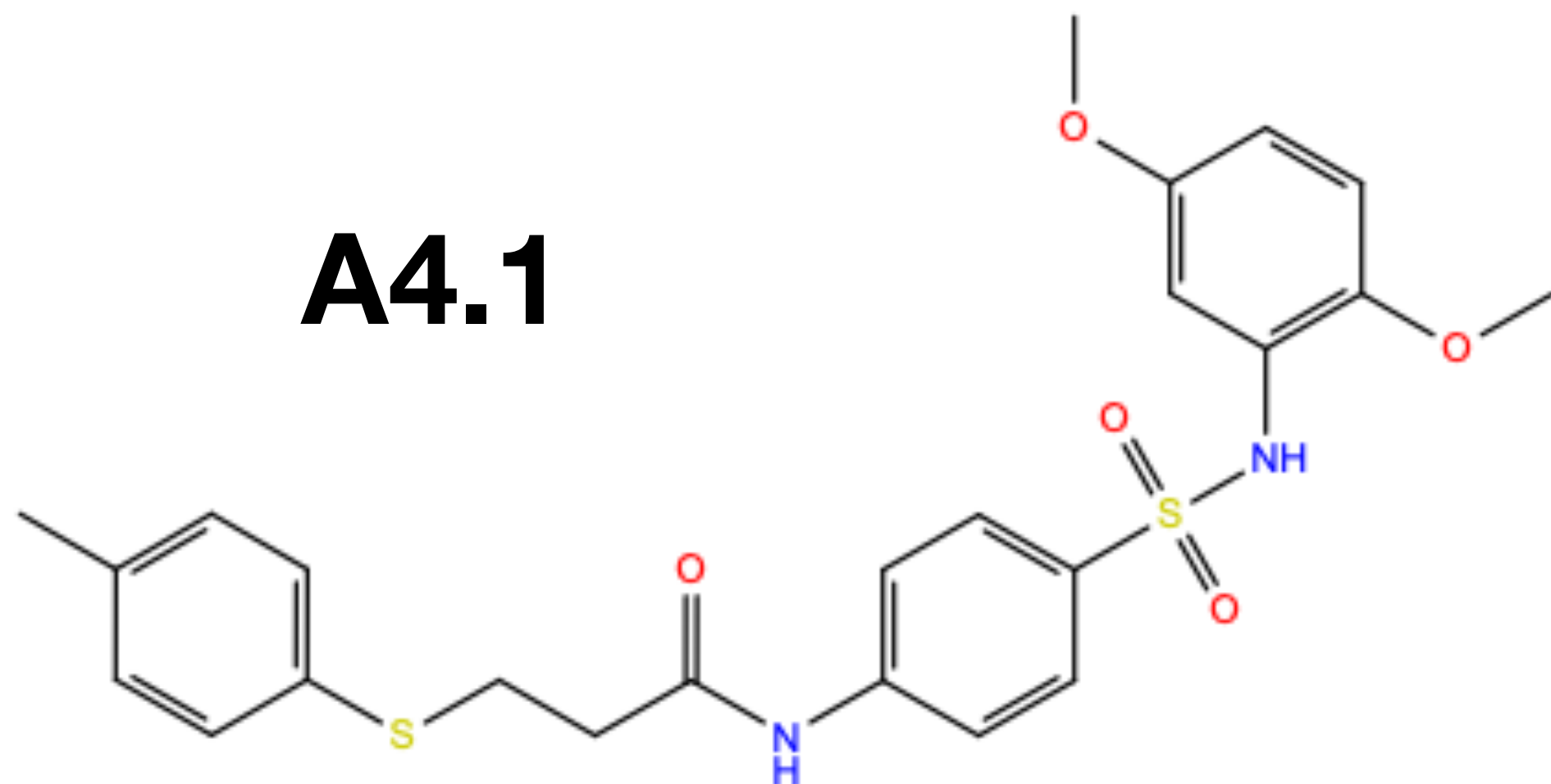
PREX1

GTP binding Site

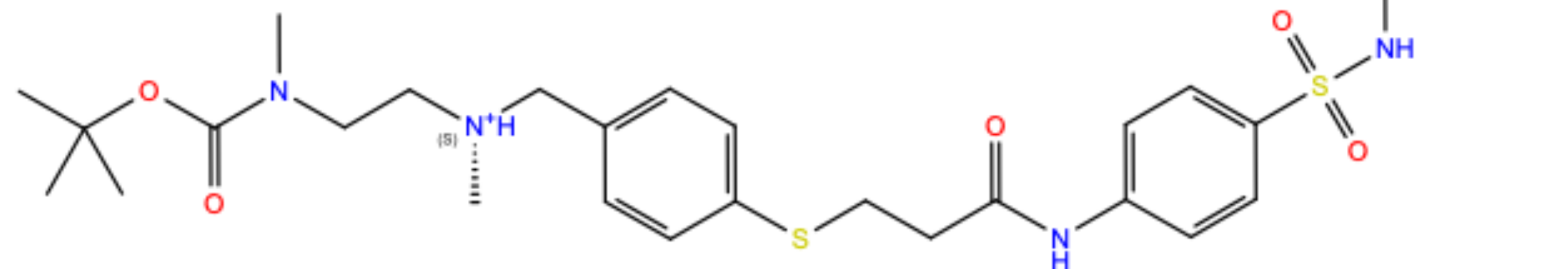
RAC1-PREX1 interface



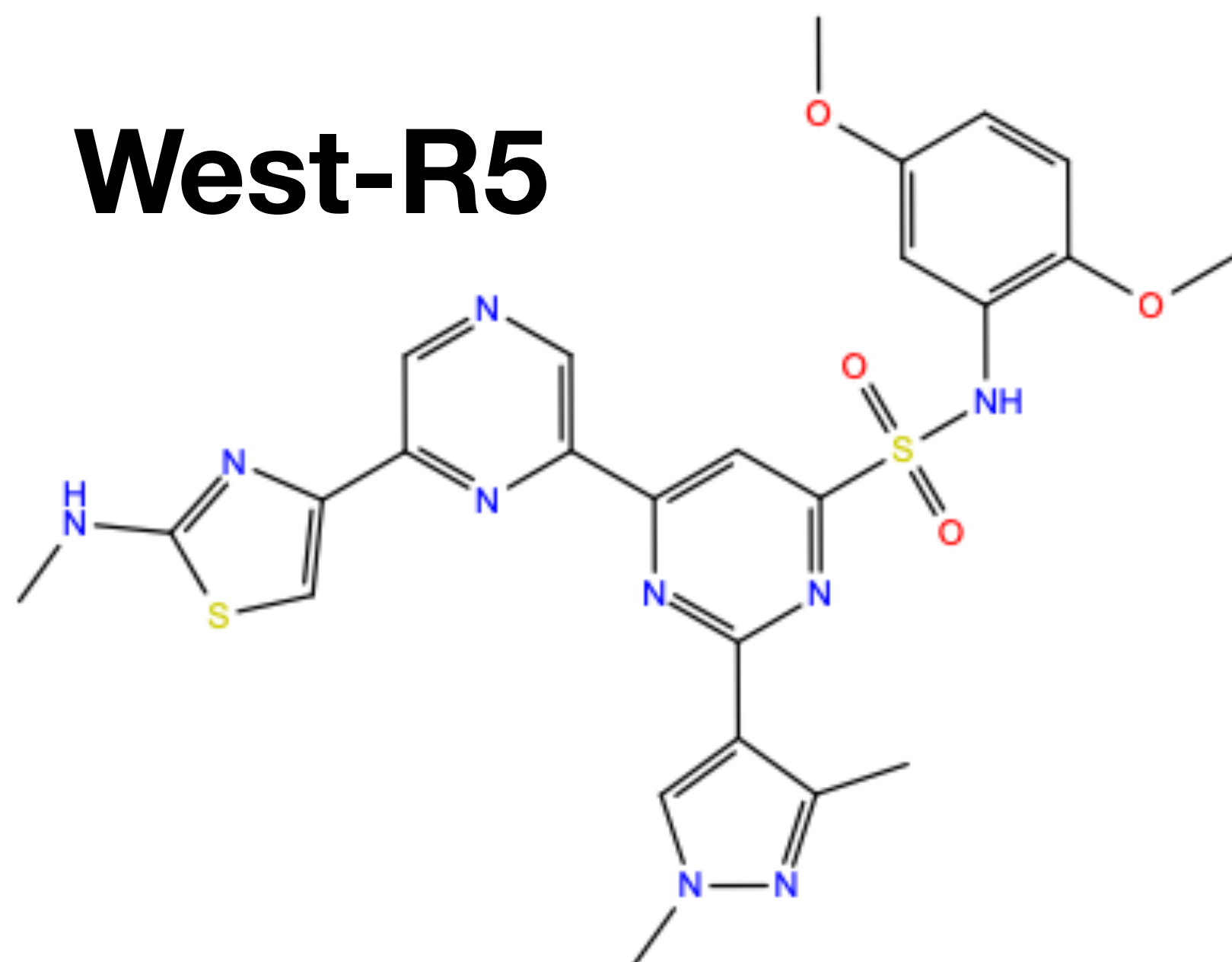
A4.1



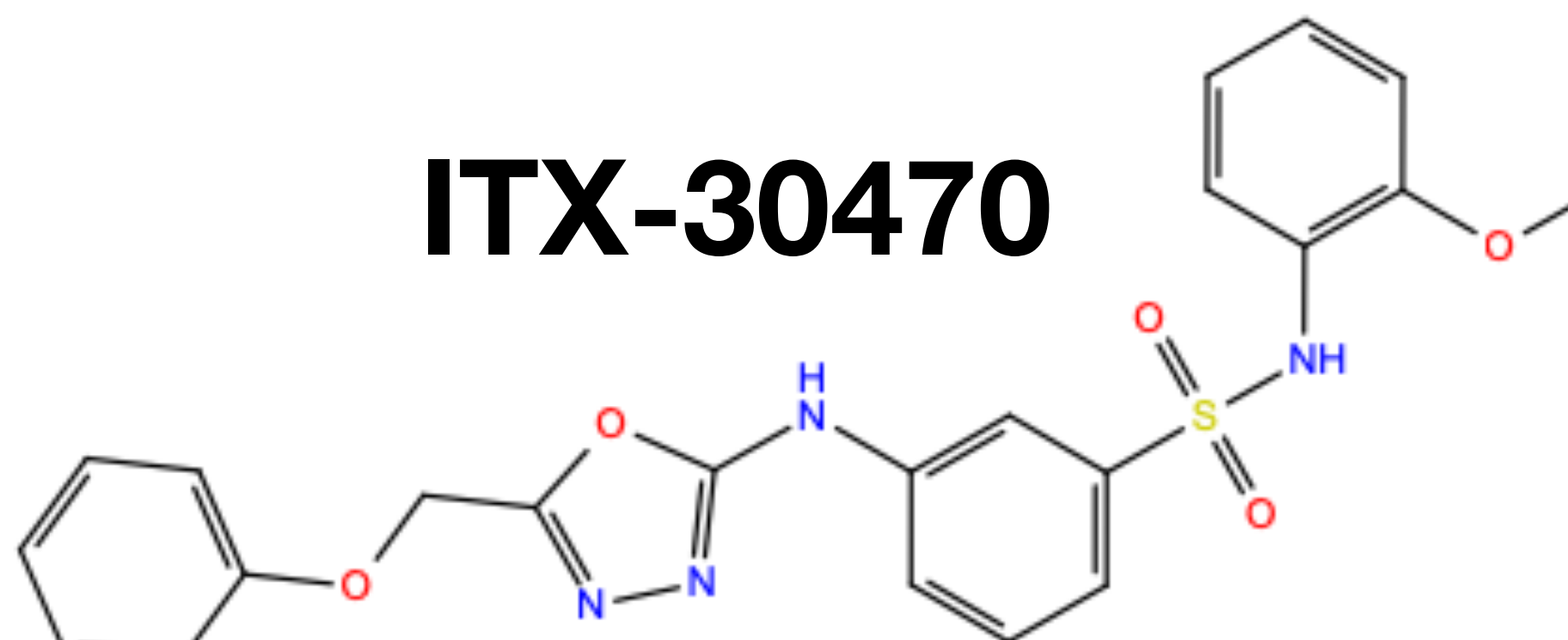
A457



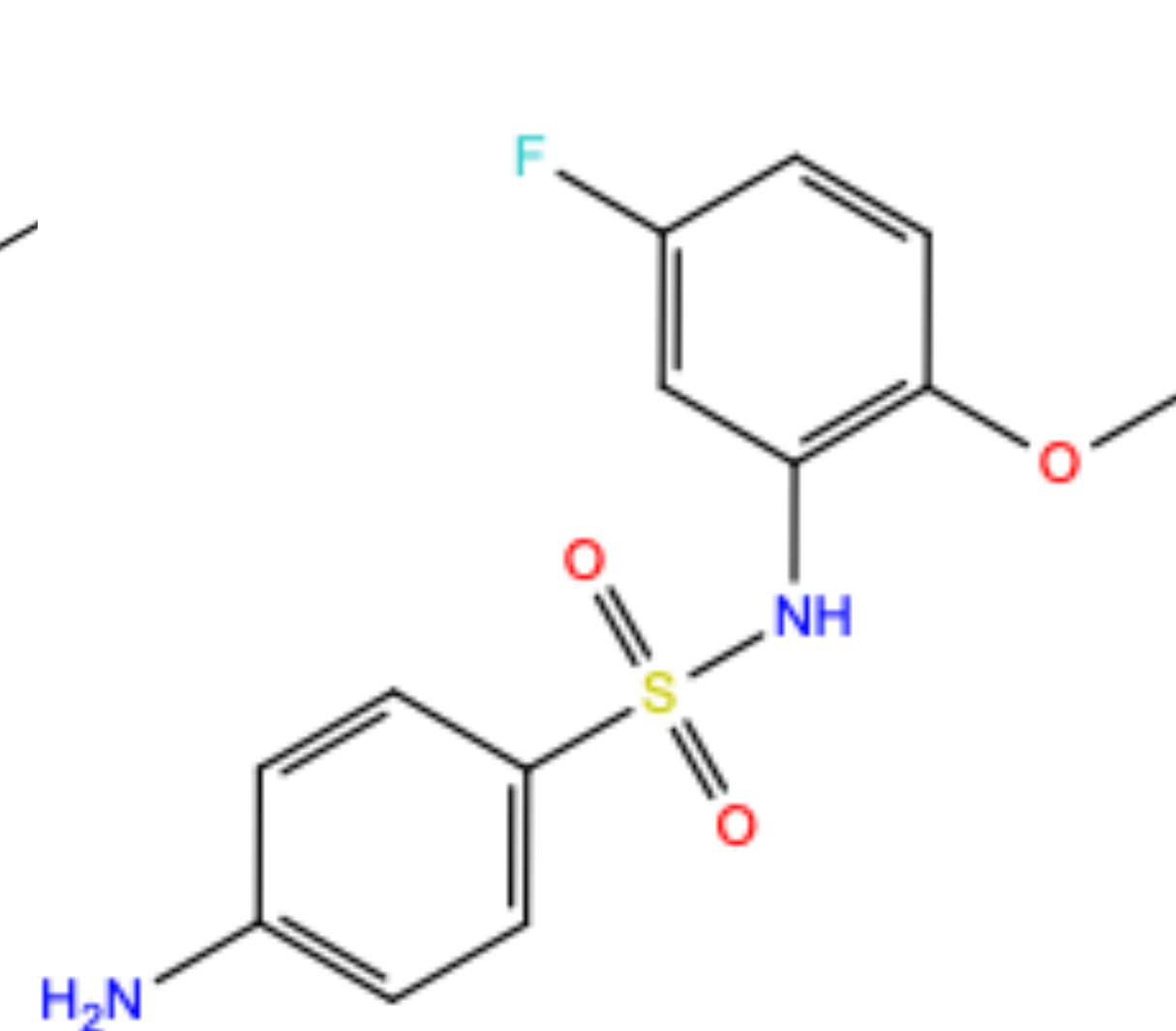
West-R5



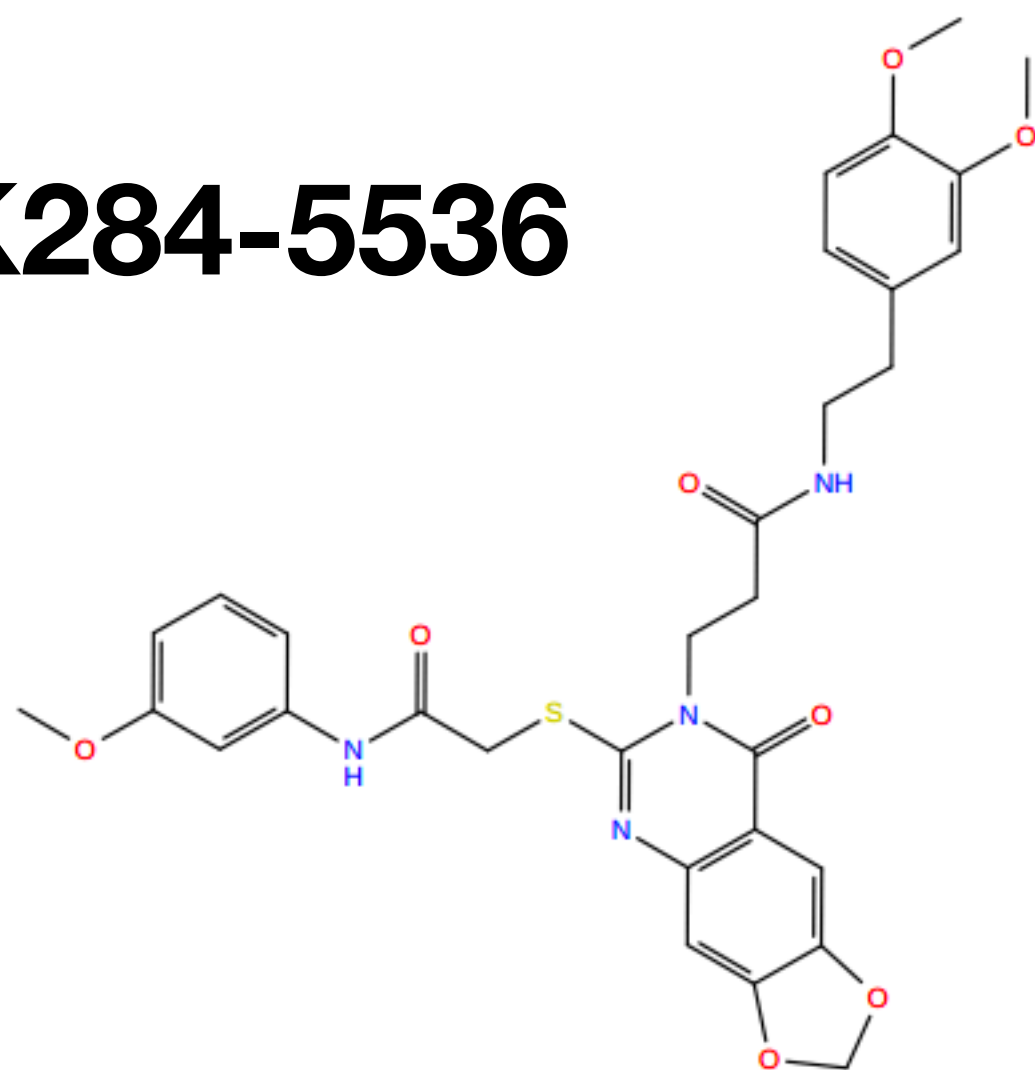
ITX-30470



ITX-30450

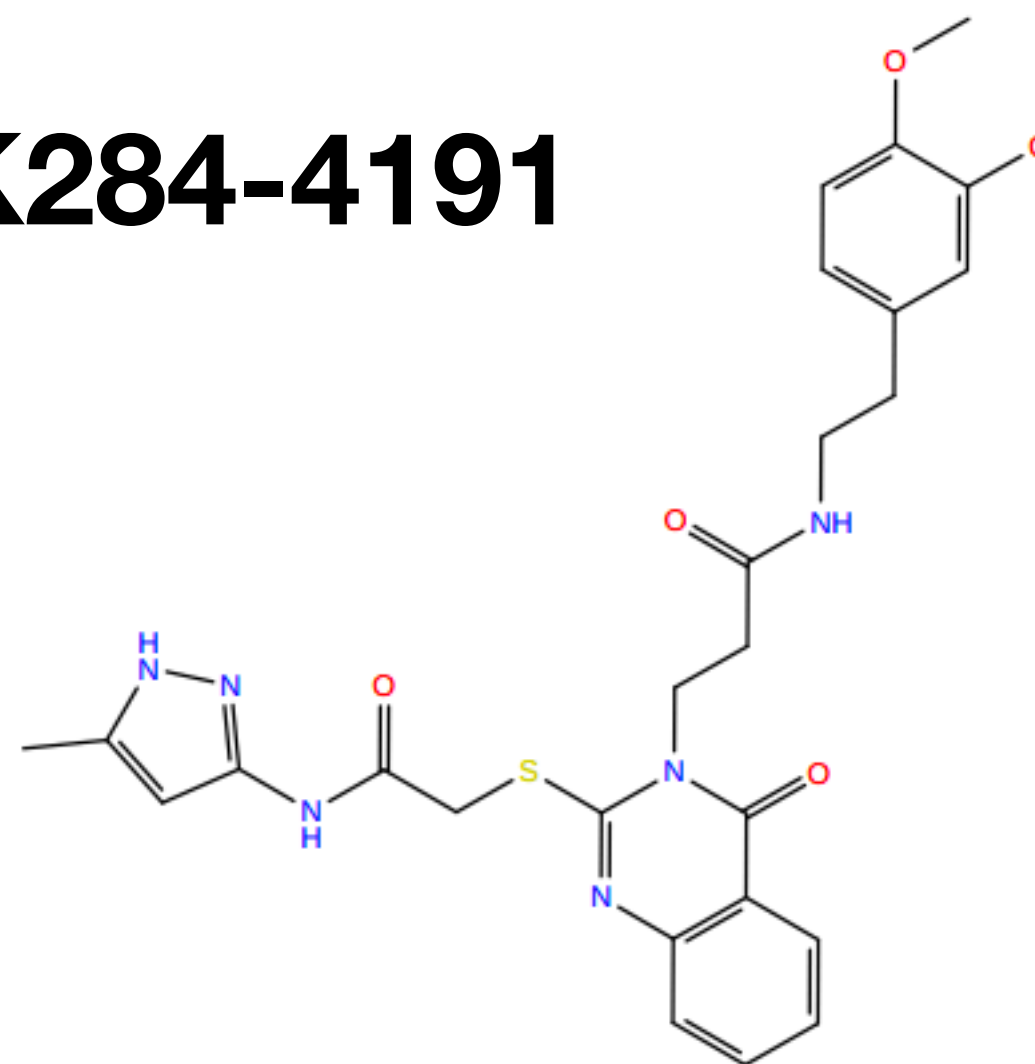


K284-5536



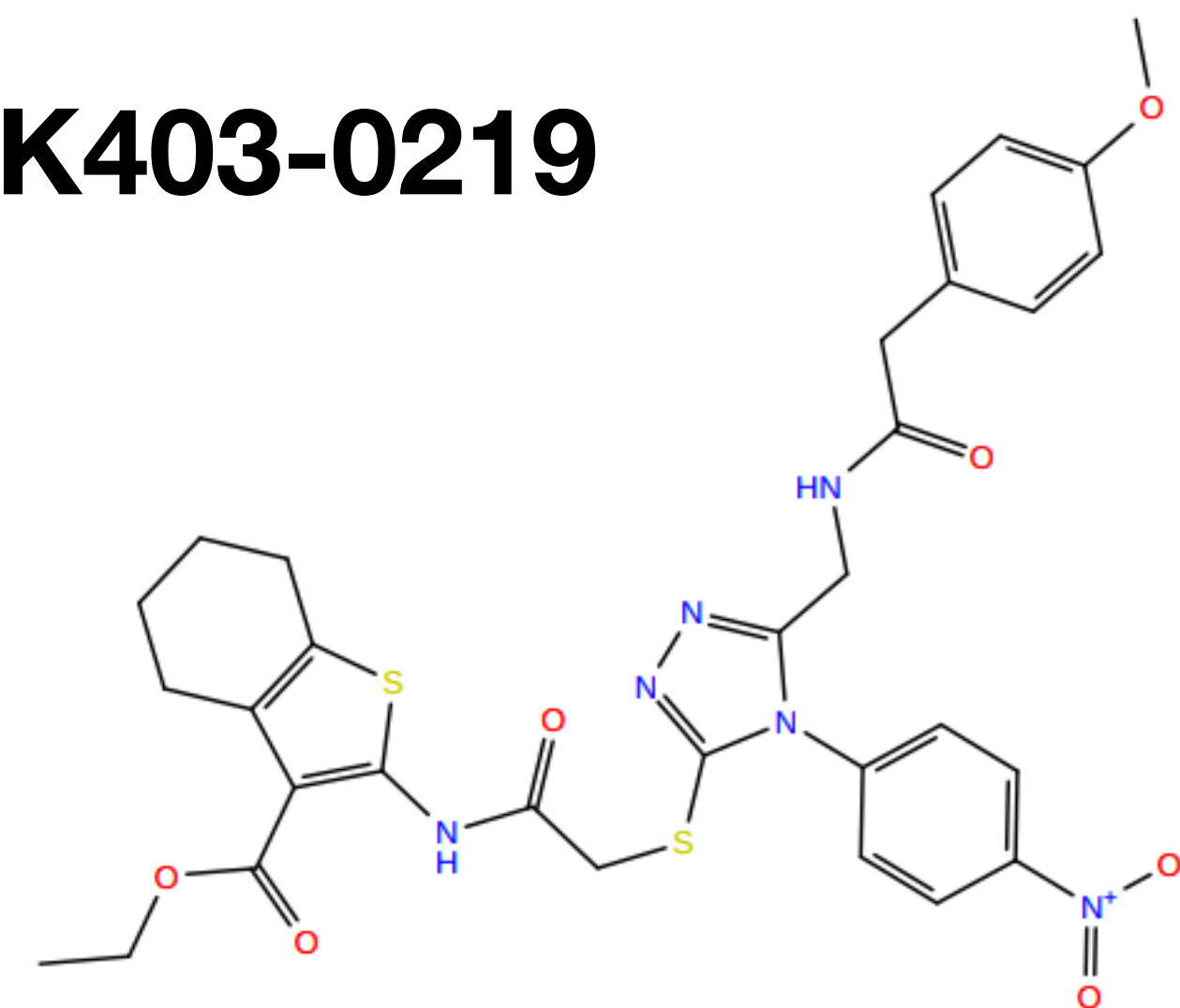
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Consensus	-148.96721
Grid Score	-88.307
LigEff-GRIDSCo	-0.142272
MMGBSA-score	-60.661
LigEff-MMGBSA	-0.097731

K284-4191



title	Chemdiv_Prote
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Grid Score	-84.297
LigEff-GRIDSCo	-0.153088
MMGBSA-score	-55.46
LigEff-MMGBSA	-0.100719

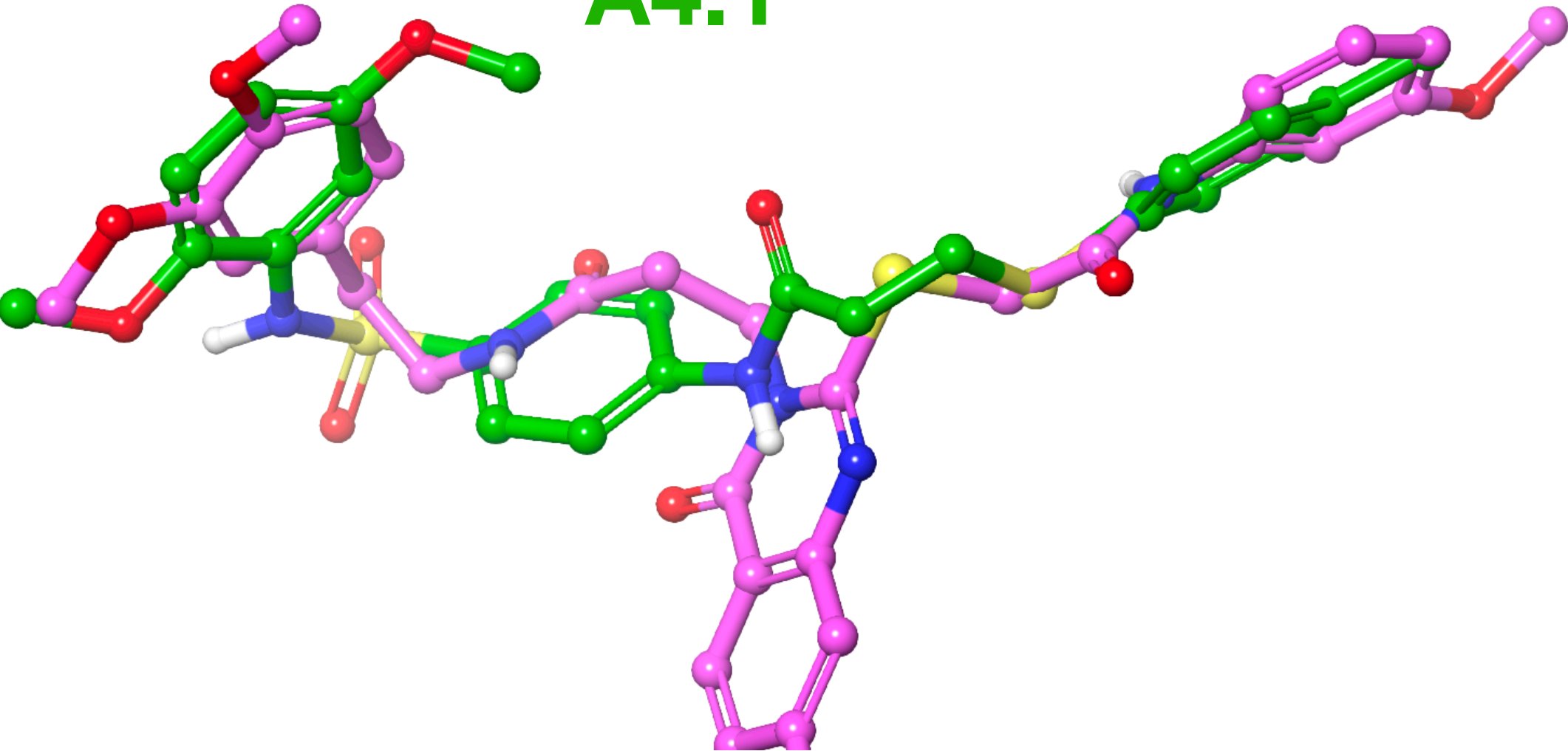
K403-0219



title	Chemdiv_Prote
Consensus	-135.21597
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LigEff-GRIDSCo	-0.128827
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LigEff-MMGBSA	-0.074577

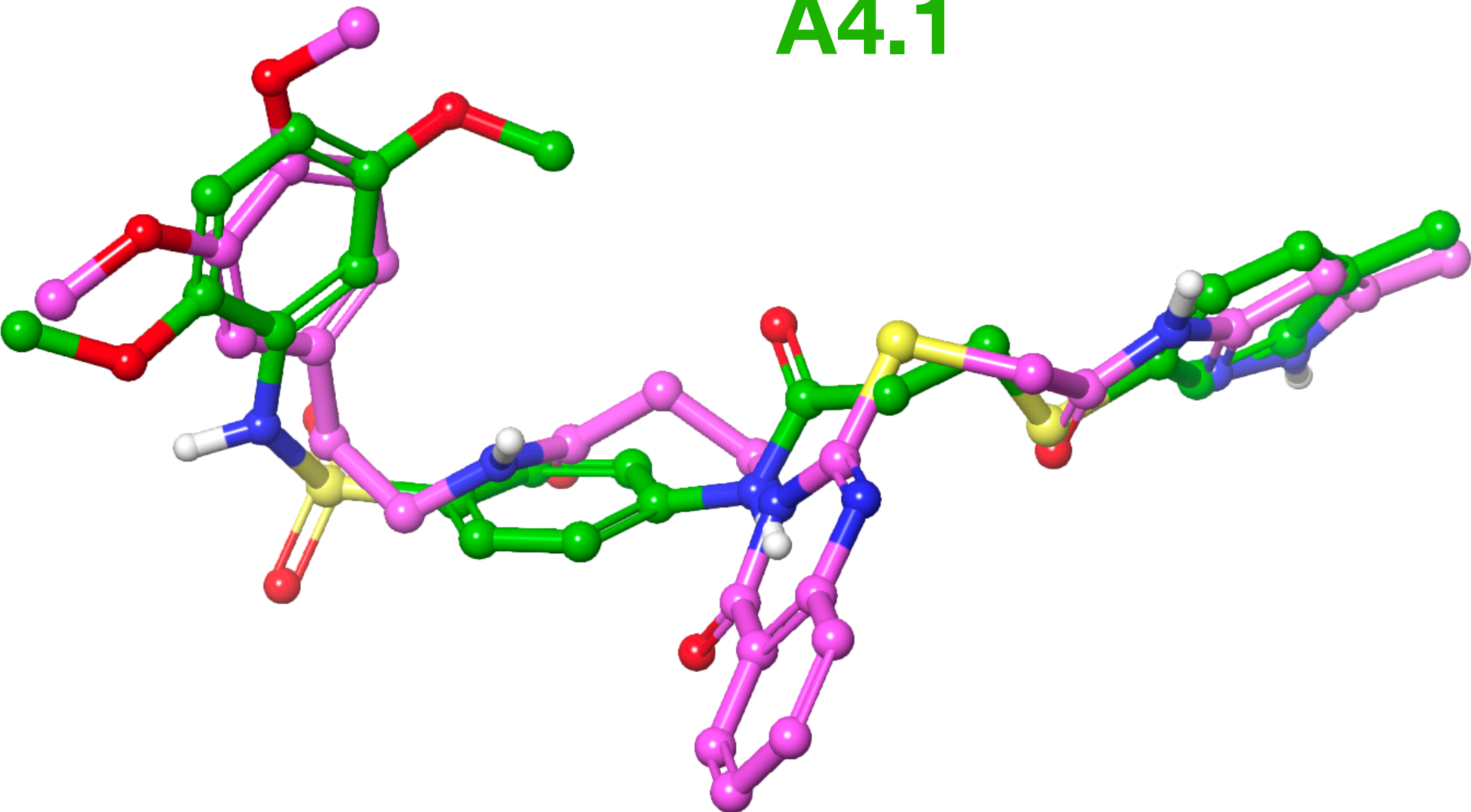
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A4.1



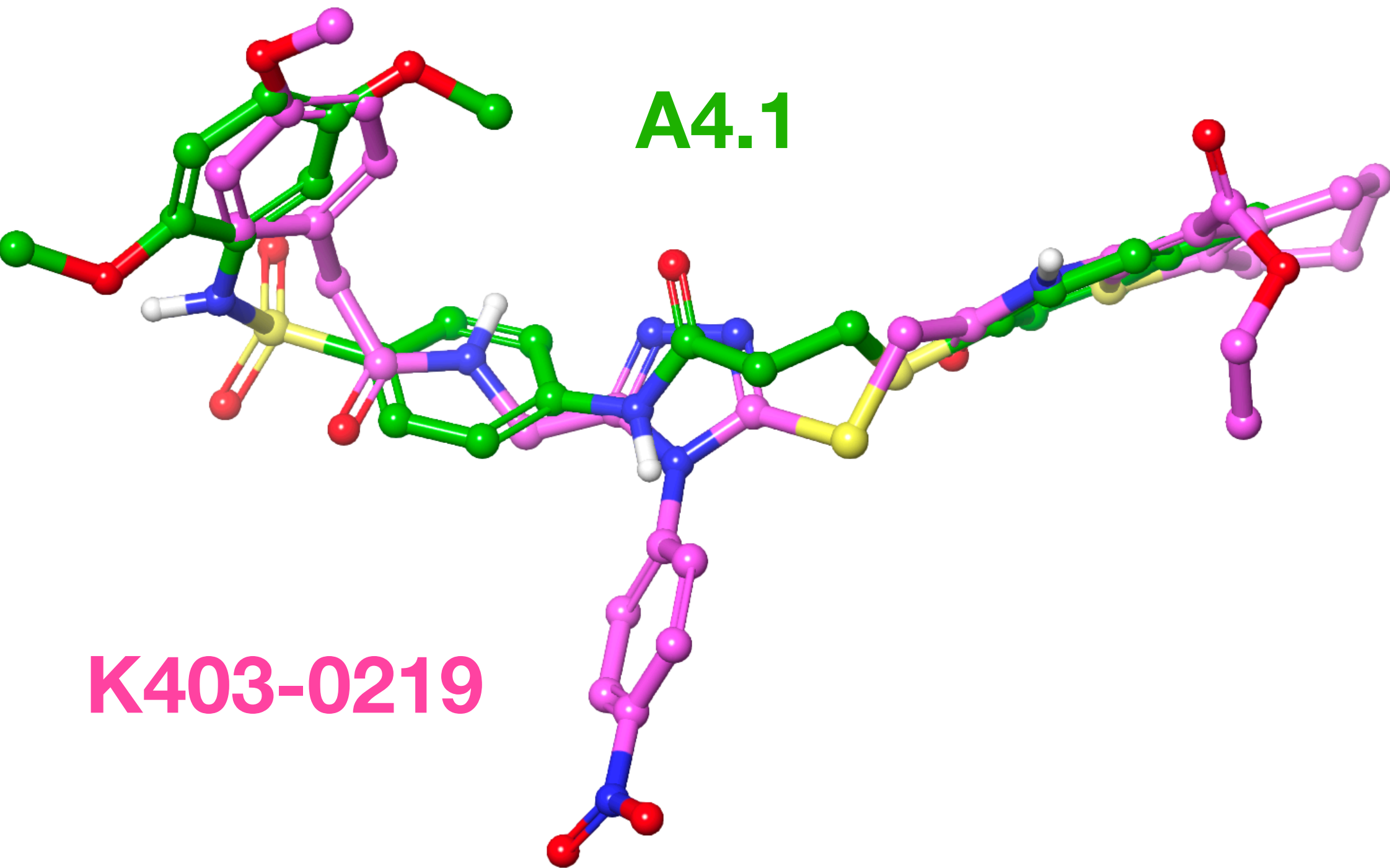
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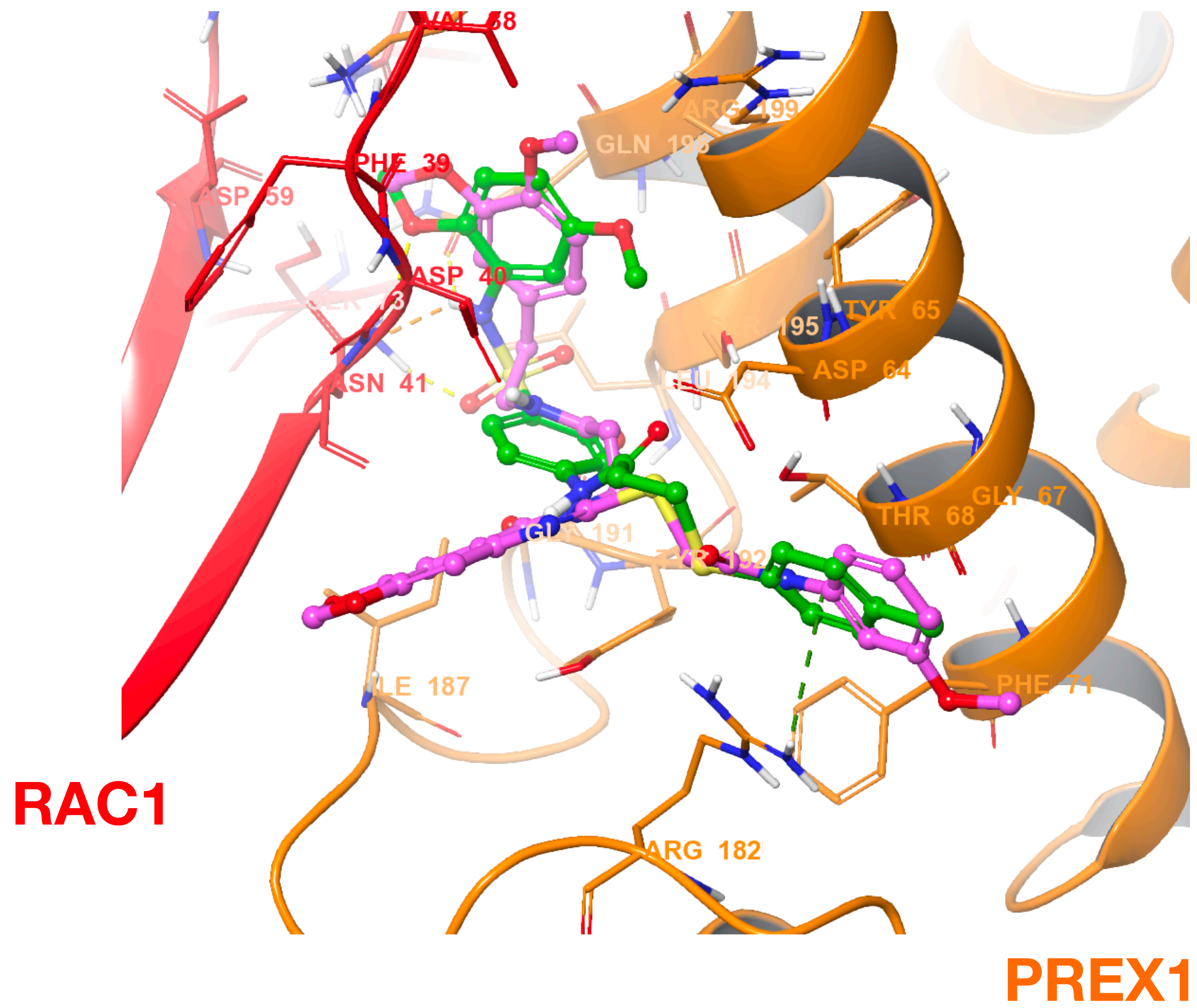
A4.1



A4.1

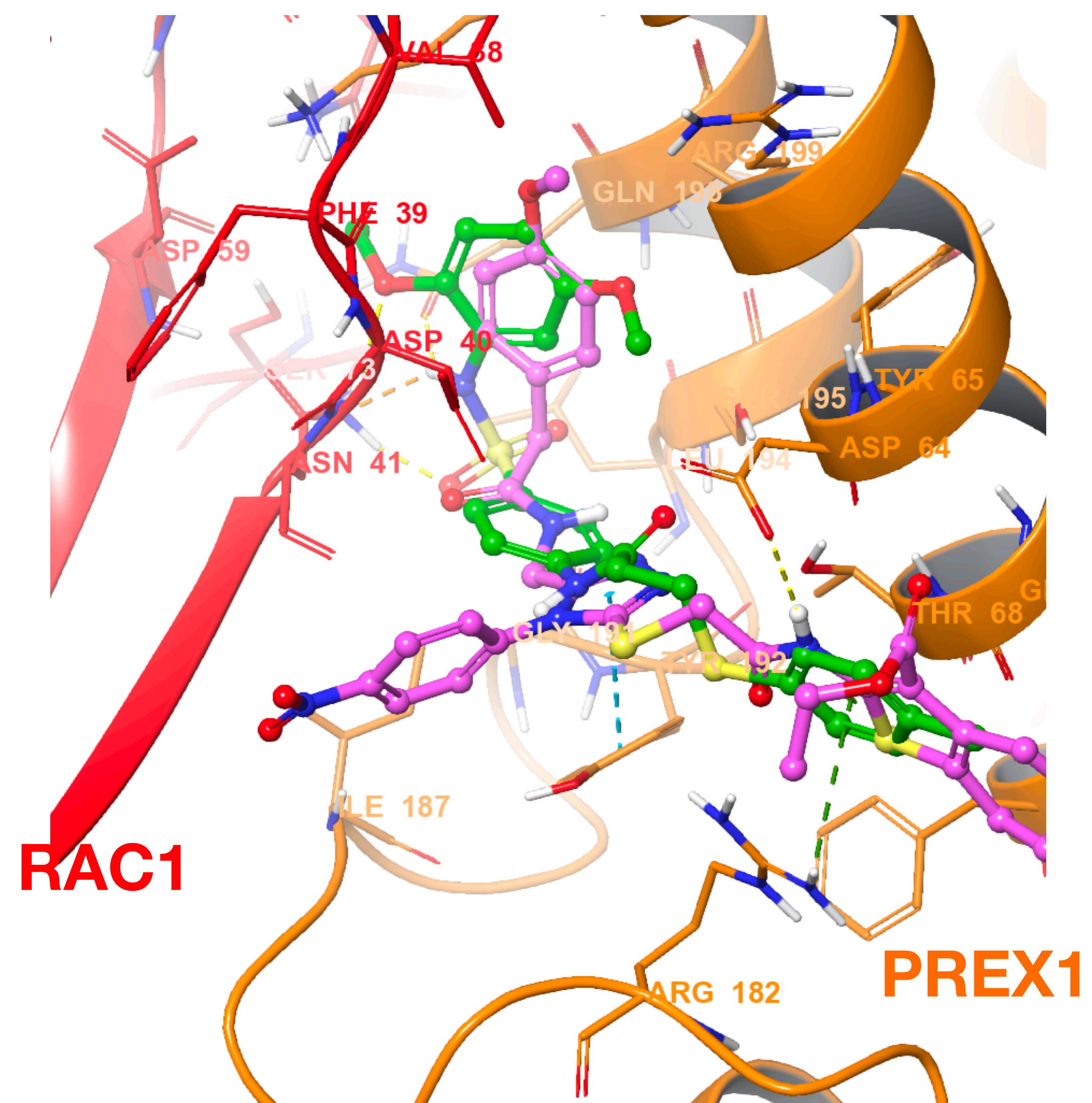
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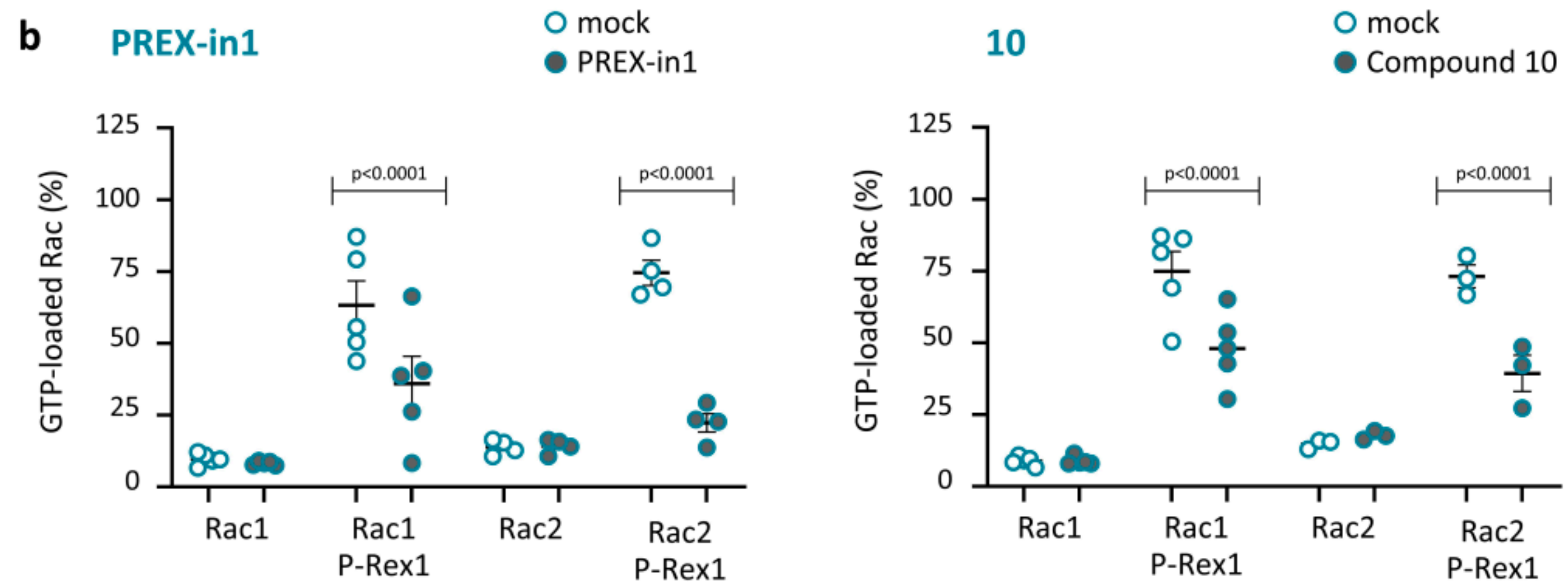
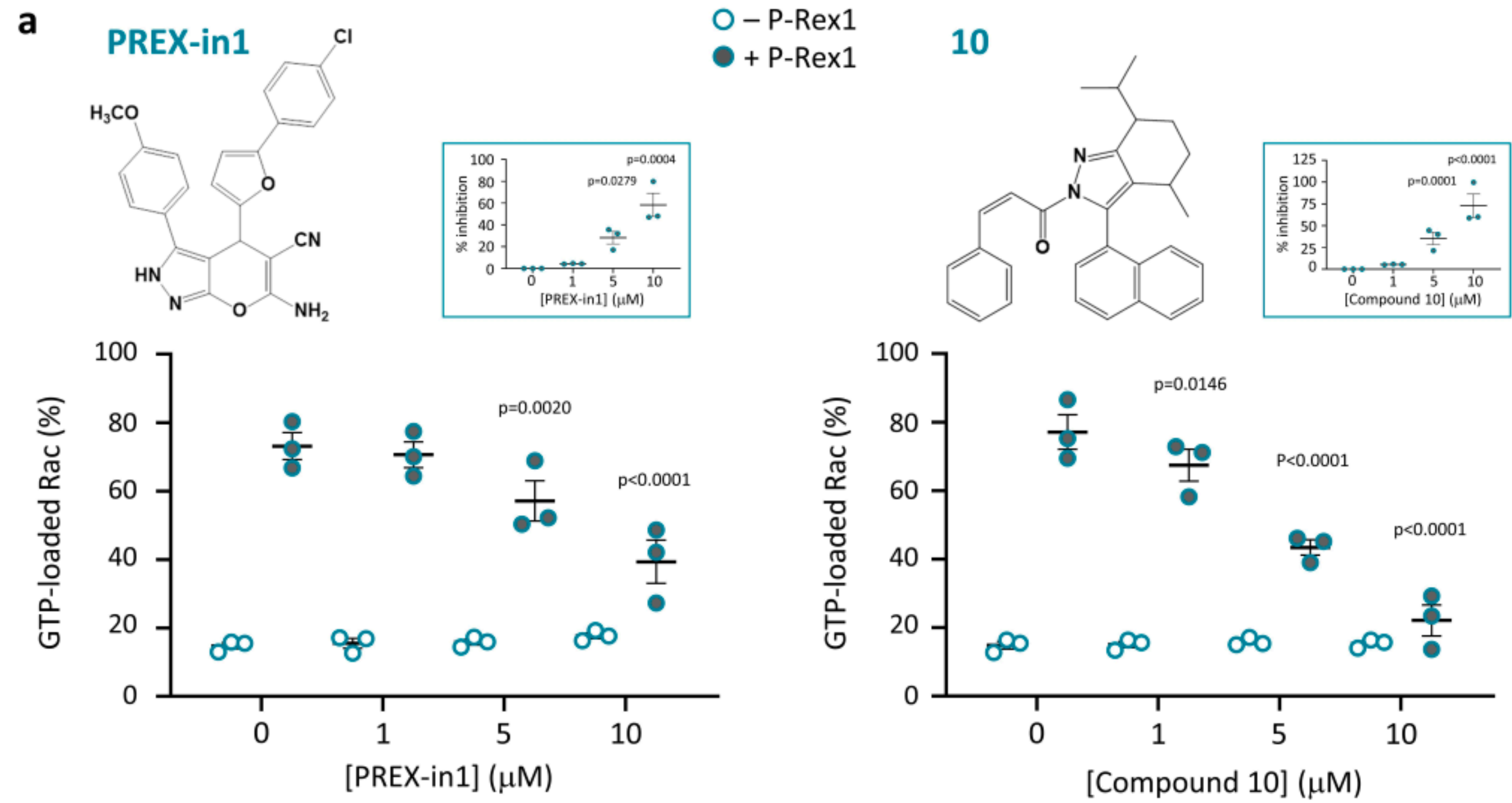
K284-5536

A4.1



K403-0219

A4.1





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Review

The Rho guanine nucleotide exchange factor P-Rex1 as a potential drug target for cancer metastasis and inflammatory diseases

Richard D. Ye

© 2014. Published by The Company of Biologists Ltd | Journal of Cell Science (2014) 127, 2589–2600 doi:10.1242/jcs.153049



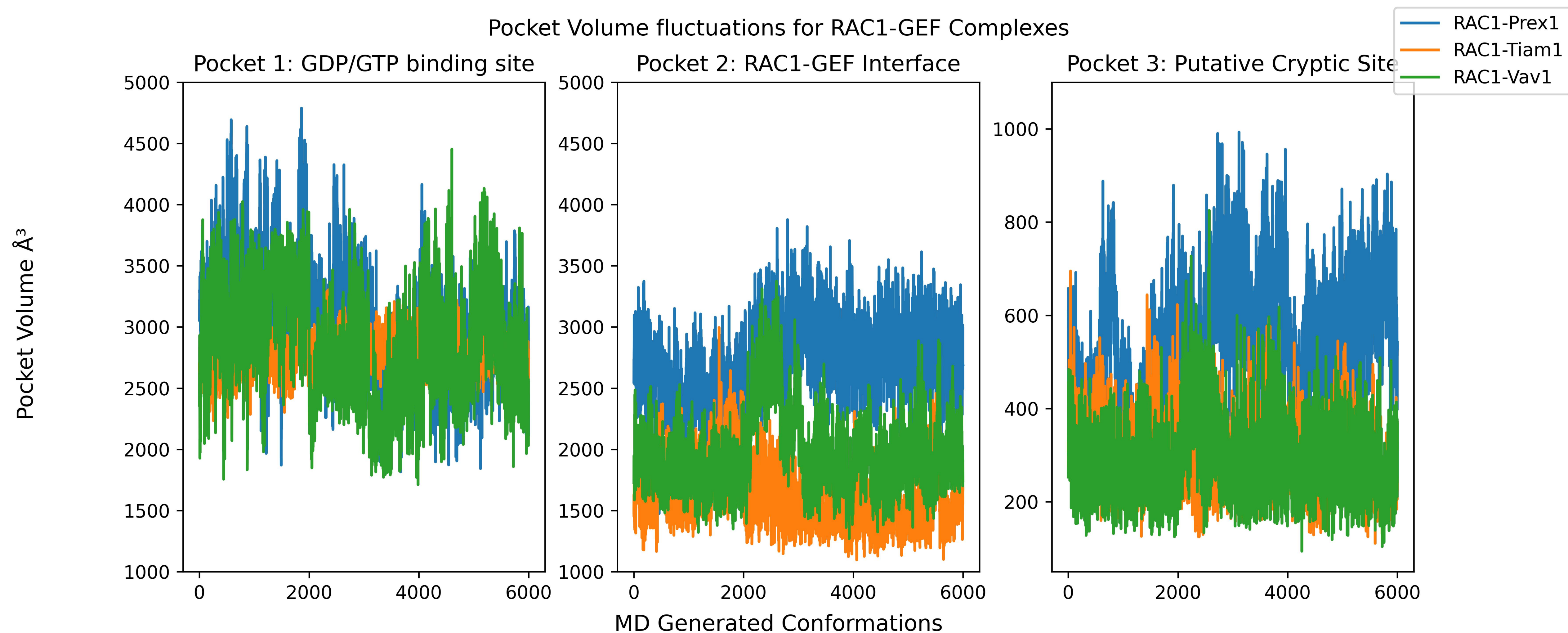
RESEARCH ARTICLE

P-Rex1 directly activates RhoG to regulate GPCR-driven Rac signalling and actin polarity in neutrophils

George Damoulakis^{1,*}, Laure Gambardella¹, Kent L. Rossman², Campbell D. Lawson¹, Karen E. Anderson¹, Yoshinori Fukui³, Heidi C. Welch¹, Channing J. Der², Len R. Stephens^{1,‡} and Phillip T. Hawkins^{1,‡,§}

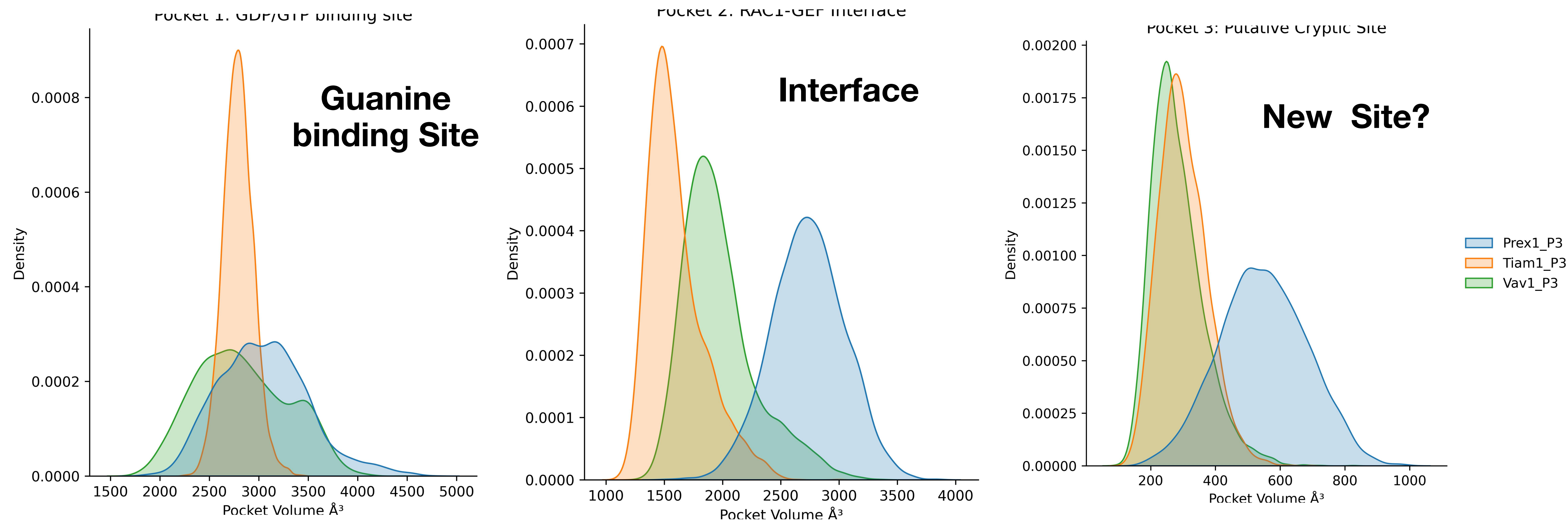
Pocket Volume Analysis: RAC1-GEFs

Time Trace of Volumes of 3 Pockets



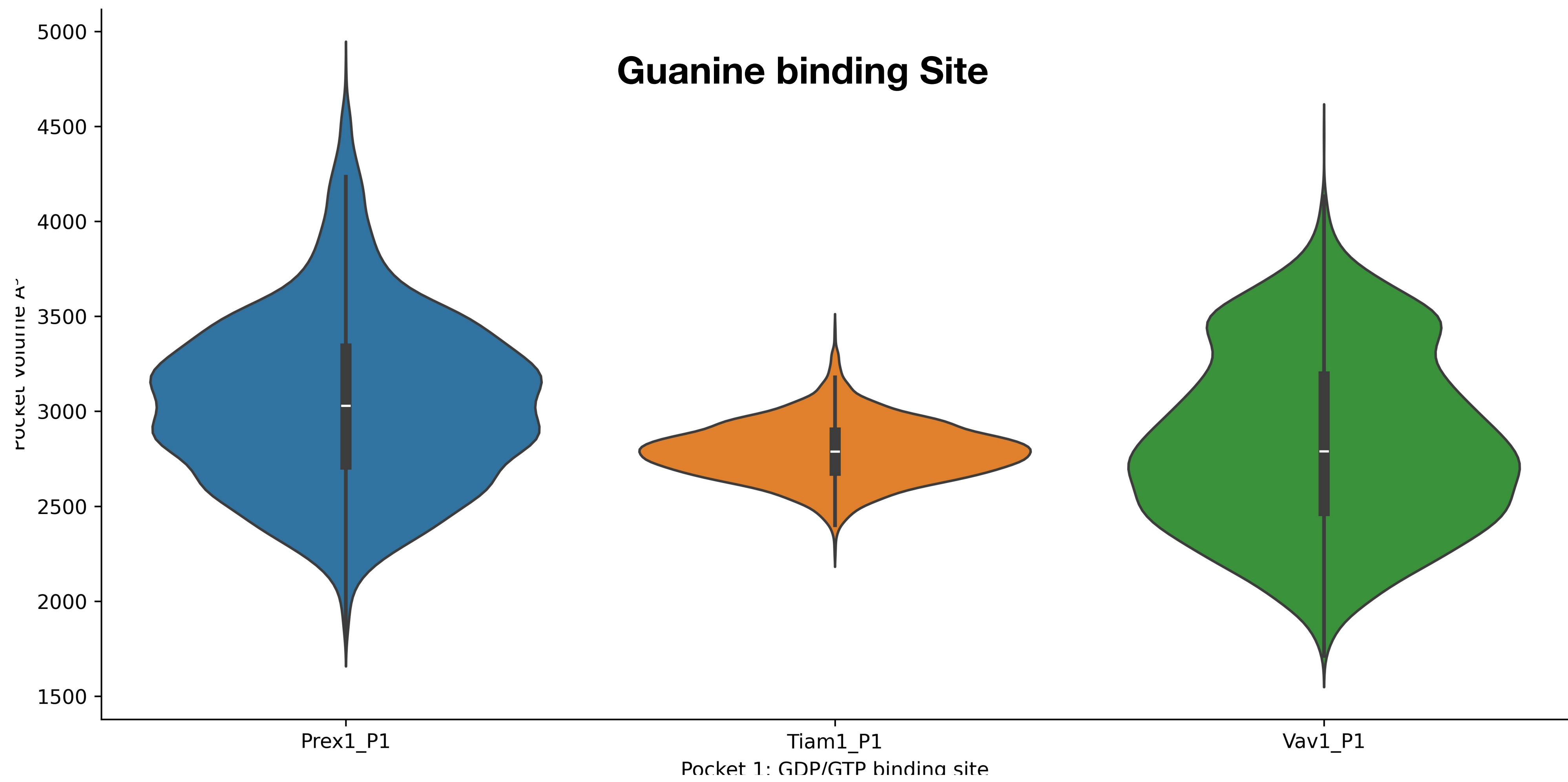
Pocket Volume Analysis: RAC1-GEFs

Central tendencies of Pocket Volumes



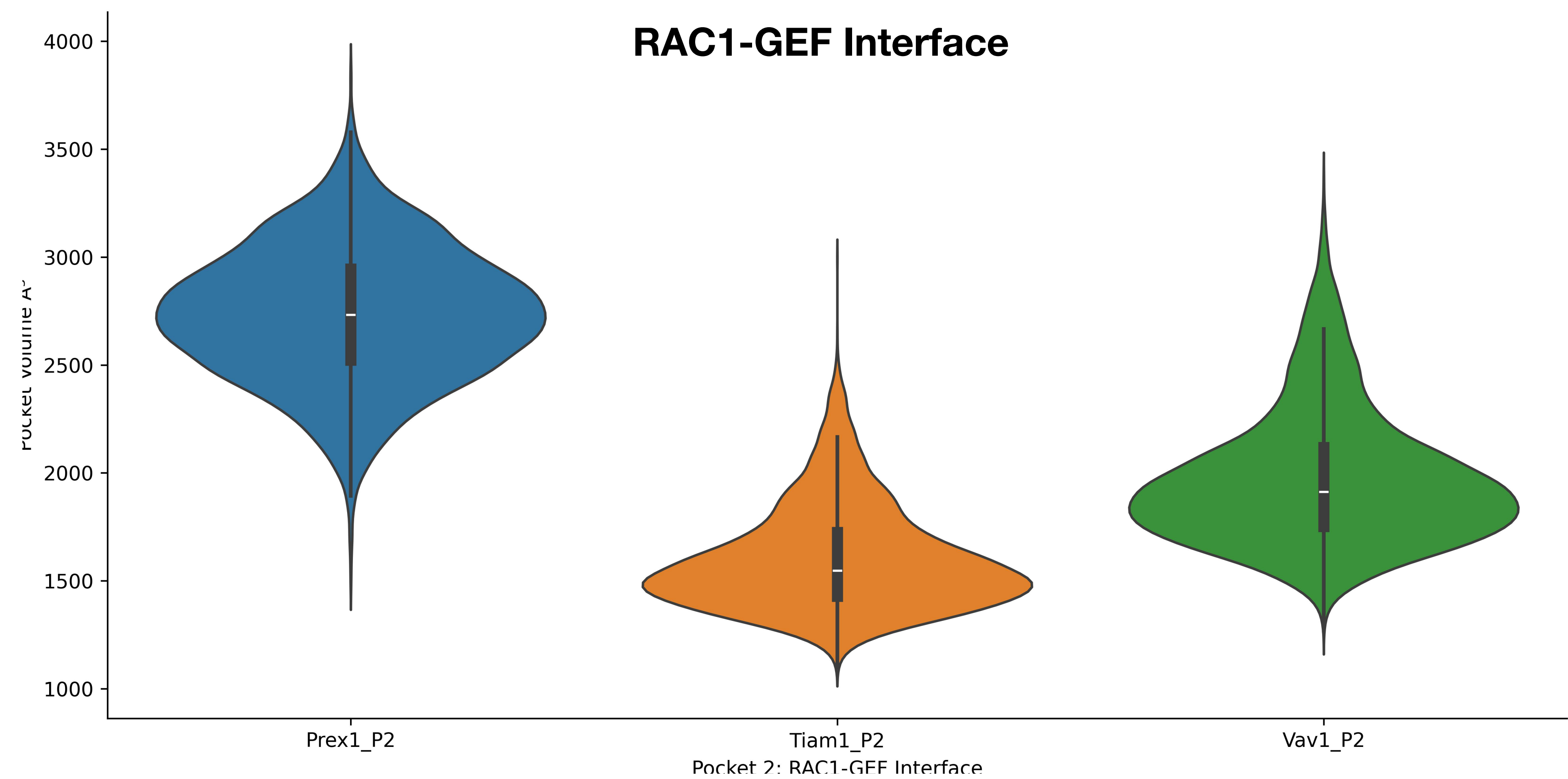
Pocket Volume Analysis: RAC1-GEFs

Central tendencies of Pocket Volumes



Pocket Volume Analysis: RAC1-GEFs

Central tendencies of Pocket Volumes



Pocket Volume Analysis: RAC1-GEFs

Central tendencies of Pocket Volumes

